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5. The following information is about the Group 7 elements of the Periodic Table, called the halogens.

Group 7 – the halogens

H																	**He**
Li	**Be**											**B**	**C**	**N**	**O**	**F**	**Ne**
Na	**Mg**											**Al**	**Si**	**P**	**S**	**Cl**	**Ar**
K	**Ca**	**Sc**	**Ti**	**V**	**Cr**	**Mn**	**Fe**	**Co**	**Ni**	**Cu**	**Zn**	**Ga**	**Ge**	**As**	**Se**	**Br**	**Kr**
Rb	**Sr**	**Y**	**Zr**	**Nb**	**Mo**	**Tc**	**Ru**	**Rh**	**Pd**	**Ag**	**Cd**	**In**	**Sn**	**Sb**	**Te**	**I**	**Xe**
Cs	**Ba**		**Hf**	**Ta**	**W**	**Re**	**Os**	**Ir**	**Pt**	**Au**	**Hg**	**Ti**	**Pb**	**Bi**	**Po**	**At**	**Rn**
Fr	**Ra**																

The Group 7 elements all exist as diatomic molecules, meaning their atoms go around in pairs. Fluorine is found at the top of the group, astatine is at the bottom of the group.

Chlorine, atomic number 17, is a pale green gas at room temperature (20 °C). It has a melting point of −101 °C. Bromine, atomic number 35, is a red-brown liquid at room temperature. It has a melting point of −7 °C. Iodine, atomic number 53, is a grey solid at room temperature. It has a melting point of 114 °C.

Chlorine is used to make bleaches, to treat public drinking water and to sterilise public swimming pools. Bromine is used by gardeners in pesticides, to prevent insects from harming plants. Iodine solution is used as an antiseptic on wounds and on patients before being operated on.

(a) Which **one** of these statements best describes the trend in the melting points of the Group 7 elements as you go down the group? Tick (✓) the correct answer. [1]

The melting point increases ☐

The melting point decreases ☐

The melting point increases and then decreases ☐

There is no trend in the melting point ☐

(b) What is the correct formula for chlorine gas? Tick (✓) the correct answer. [1]

CL2		Cl_2		cl_2		2Cl	

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(c) Predict a value for the boiling point of chlorine and explain your answer. [2]

Boiling point of chlorine .... °C

Explanation ....

....

(d) Give the property that the Group 7 elements have that means they can be used as described in the text. [1]

....

(e) Decide whether the following statements about astatine are true or false. [2]

Astatine ...	True / False
... is a solid at room temperature	
... conducts electricity	
... has a melting point higher than 114 °C	
... is coloured	

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